

Principles of Discrete Applied Mathematics

Personal Notes

Mihir Rao

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1 The Pigeonhole Principle

The *pigeonhole principle* is among the simplest statements in mathematics and among the most surprisingly powerful. It says that objects cannot be distributed among fewer containers without doubling up somewhere, yet it settles questions that look nothing like counting problems. This lecture applies it to graphs, subset sums, monotone subsequences, and the information-theoretic limits of a guessing game.

Begin with a puzzle. A mathematician gets up in the dark and, to avoid waking their partner, dresses in another room. They grab socks from a drawer containing only two colors. How many socks guarantee a matching pair?

The answer is three. Two socks may be mismatched, but among three socks and only two colors, some color must appear twice. The general argument is already visible: when there are more socks than colors, two socks share a color.

1.1 Statement

Definition: If m objects, called pigeons, are placed into n boxes, called pigeonholes, and $m > n$, then at least one box contains more than one object.

A stronger form gives a quantitative guarantee rather than a bare existence claim.

Theorem 1.1 [*Pigeonhole principle, strong form*]. If m objects are placed into n boxes, then at least one box contains at least

$$\left\lceil \frac{m}{n} \right\rceil$$

objects.

Here $\lceil \cdot \rceil$ denotes rounding up to the next integer, while $\lfloor \cdot \rfloor$, used later, denotes rounding down. The basic form is the case $m > n$, where $\lceil m/n \rceil \geq 2$.

Note. The principle is so elementary that it is natural to suspect nothing interesting can follow from it. That suspicion is wrong. The difficulty is never the principle itself; it is choosing what to call a pigeon and what to call a pigeonhole.

1.2 Two Vertices of Equal Degree

A *graph* G consists of a set V , whose elements are called *vertices*, and a set E of pairs of vertices, whose elements are called *edges*. The *degree* of a vertex v is the number of edges containing v .

Theorem 1.2. In any finite graph, there are two vertices of equal degree.

At a party with n people, this says that two people know the same number of people at the party.

Proof. Let the graph have n vertices. Every degree is an integer between 0 and $n - 1$, which gives n possible values for n vertices, so counting alone does not force a repeat. Suppose for contradiction that all degrees are distinct. Then every possible value occurs exactly once. In particular, some vertex v has degree 0, while some vertex w has degree $n - 1$ and is adjacent to every other vertex.

These two vertices cannot coexist. If the edge (v, w) is present, then v has a neighbor and does not have degree 0. If the edge is absent, then w is not adjacent to every other vertex and does not have degree $n - 1$. Either case is a contradiction, so two degrees must coincide. $\square$

Note. The pigeons are the n vertices and the pigeonholes are the n possible degrees, so the counts initially match. The proof rules out simultaneous use of degrees 0 and $n - 1$, leaving only $n - 1$ usable holes for n pigeons.

1.3 Equal Sum Subsets

Here the principle does something less obvious.

Theorem 1.3. *Among any 30 seven-digit numbers, there exist two disjoint nonempty subsets with the same sum.*

Proof. Take the nonempty subsets as pigeons. There are $2^{30} - 1$ of them. Take the possible subset sums as pigeonholes. Each number is less than 10^7 , so the sum of at most thirty numbers is at most $3 \cdot 10^8$, which is smaller than $2^{30} - 1 \approx 10^9$. There are therefore more subsets than achievable sums, so two distinct subsets A and B satisfy

$$\sum_{x \in A} x = \sum_{x \in B} x.$$

The subsets need not initially be disjoint, but their common elements can be discarded. Put $A' = A \setminus B$ and $B' = B \setminus A$. Removing the same quantity $\sum_{x \in A \cap B} x$ from both sides preserves equality, while A' and B' are disjoint. Since A and B were distinct, at least one of A' and B' is nonempty; equality of sums forces the other to be nonempty as well. $\square$

Note. The proof establishes existence without providing an efficient construction. At larger scales, finding the subsets can require enormous computation even though their existence is immediate. Pigeonhole arguments naturally separate existence from construction.

1.4 Monotone Subsequences

A *subsequence* of $a_1, a_2, \dots, a_k$ is obtained by deleting some terms while preserving the order of the rest. It is *monotone* when it is entirely increasing or entirely decreasing. Long sequences of distinct real numbers must contain long monotone subsequences.

Consider the length-12 sequence

$$4, 3, 2, 1, 8, 7, 6, 5, 12, 11, 10, 9.$$

The terms 4, 8, 11 form an increasing subsequence of length 3, while 4, 3, 2, 1 form a decreasing subsequence of length 4. Neither can be improved within this sequence. With $m = 3$ and $n = 4$, it has exactly $mn = 12$ terms, one short of the $mn + 1 = 13$ terms that force a longer subsequence.

Theorem 1.4. *Any sequence of $mn + 1$ distinct real numbers $a_1, a_2, \dots, a_{mn+1}$ has an increasing subsequence of length $m + 1$ or a decreasing subsequence of length $n + 1$.*

Proof. Let $t(i)$ denote the length of the longest increasing subsequence ending at a_i . If $t(i) > m$ for some i , an increasing subsequence of length at least $m + 1$ exists and we are done. Assume instead that $t(i) \in \{1, 2, \dots, m\}$ for every i .

Apply the pigeonhole principle to the $mn + 1$ indices and the m possible values of t . Some value s satisfies $t(i_j) = s$ for at least $n + 1$ indices

$$i_1 < i_2 < \dots < i_{n+1}.$$

We claim that

$$a_{i_1} > a_{i_2} > \dots > a_{i_{n+1}},$$

which is a decreasing subsequence of length $n + 1$. Otherwise some consecutive pair satisfies $a_{i_j} < a_{i_{j+1}}$. The increasing subsequence of length s ending at a_{i_j} could then be extended by $a_{i_{j+1}}$, producing an increasing subsequence of length $s + 1$ ending there. This contradicts $t(i_{j+1}) = s$. $\square$

Exercise. For any m and n , construct a sequence of mn distinct numbers with no increasing subsequence of length $m + 1$ and no decreasing subsequence of length $n + 1$. This shows the theorem is best possible.

1.5 Twenty Questions

One player thinks of an object and a second player asks yes-or-no questions until they can name it. How many distinct objects can be distinguished with twenty questions?

The strategy may be adaptive: the k th question can depend on the preceding $k - 1$ answers. Restrict

the guesser to a deterministic strategy, so that the same answer history always produces the same next question.

Even with adaptivity, the guesser must distinguish every pair of objects. If two objects produced the same answers throughout, the strategy would ask the same questions for both and finish in the same state, forcing an incorrect guess for at least one of them.

Thus each object determines a unique sequence of twenty yes-or-no answers. Take objects as pigeons and answer sequences as pigeonholes. There are 2^{20} sequences, so at most 2^{20} objects, a little over one million, can be distinguished. This bound is attained by a strategy that separates 2^{20} objects in twenty questions.

Note. Allowing a probabilistic strategy does not increase the number of objects that can be distinguished, though proving this requires probability not yet developed in the course.

1.6 Twenty Questions with One Lie

Now allow the answerer to lie once, a variation apparently first considered by Stanislaw Ulam. How many objects can the guesser distinguish?

For a fixed object, the answerer may tell the truth throughout or lie on any one of the twenty questions. Once the lie is used, all remaining answers are forced. Hence each object is compatible with 21 distinct answer sequences. With t objects there are $21t$ pigeons and still 2^{20} pigeonholes.

If $21t > 2^{20}$, two pigeons share a hole. They cannot represent the same object because its 21 compatible sequences are distinct. Therefore some answer sequence is compatible with two objects, so the guesser cannot distinguish them. The maximum is bounded by

$$t \leq \left\lfloor \frac{2^{20}}{21} \right\rfloor = 49932.$$

Note. This is an upper bound; whether a strategy attains it is a separate question. For some numbers of questions such a strategy exists, and otherwise one can come reasonably close. The relevant tools appear later with error-correcting codes: a lie is a transmission error, and the questions must be robust to one such error.

Source: MIT 18.200 Principles of Discrete Applied Mathematics, Spring 2024, [Lecture 1 notes](#) by Ankur Moitra.